

Multimodal Traffic Analysis and Accident Prevention System

Problem Statement:

This project aims to develop an AI-based traffic monitoring and accident prediction system using text, image, audio, and video data. The system analyses traffic conditions, identifies accident-prone areas, predicts possible accident risks, and determines traffic density as low, medium, or high.

The system uses:

- Video data to analyse vehicle movement, traffic congestion, and over speeding.
- Image data to detect road damage, potholes, and narrow roads.
- Audio data to identify abnormal traffic sounds such as excessive horns or collision sounds.
- Text data from reports and traffic updates to understand traffic causes and accident patterns.

The system also studies:

- when traffic becomes high,
- how long traffic lasts,
- whether traffic clears automatically,
- and whether police control or any preventive measure is required.

The main objective of the project is to improve traffic management and reduce accident risks using Machine Learning and Deep Learning techniques.